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The Editor-in-Chief
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Dear Editor,

I submit the manuscript *Certified reliability for any routing solver: screening buffered plans against travel-time uncertainty* for consideration as a research article.

Delivery firms promise time windows, but most plan with fixed travel times and protect themselves by padding every trip, which says nothing about how often the plan will be on time. Methods that model the uncertainty directly need a solver built for that model. The manuscript takes the systems view of a firm that keeps its existing solver and adds a reliability layer around it: the solver produces plans with capped internal buffers, the plans are screened by simulation, and the chosen plan is certified on independent simulated days.

The contributions are:

- a reliability layer that works with an unchanged solver, with a certificate that, under the modelled delays, holds for any solver and selection rule (Theorem 1). The guarantee applies a standard a posteriori result from sampled chance-constrained programming; the contribution is the layer and the evidence, not new statistical theory. Two further results say when a certificate survives a wrong delay model: it carries over to any real delays no worse than modelled (Proposition 1), and certifying against a set of models guarantees each of them (Proposition 2);
- evidence that this works: none of 666 certified plans failed its certificate on fresh days. Certificates under one model often failed under harsher delays; certifying against five models removed every failure, at a cost the paper reports in full: a plan was certified on 76 of 159 instances instead of 131, with 0.65 more vehicles where both screens selected one;
- an ablation with the HGS solver showing that the gain comes from the buffers and the screen, not from the planner's objective: capped buffers planned for distance alone and screened were 28.3 points more reliable than 20% padding, as reliable as the full procedure, while the weighted plan without buffers or screening was less reliable than padding;
- fixed-order certification (Proposition 4), which lets a firm certify the cheapest good plan instead of the screened one with the same guarantee; a certified slowdown margin (Proposition 5) that says how much slower than modelled every trip may be before a certificate breaks, which shows that most certificates were fragile and that those with a margin of 10% or more survived every harsher delay model; and a count of how many real days of history a certificate without a delay model needs (59 at 95%, Proposition 3);
- tests beyond synthetic data: benchmark instances; real Amazon delivery routes with delays fitted to 6.2 million New York taxi trips, an illustration on real routes with assumed windows; and real pickup operations with real windows and GPS tracks in five Chinese cities, which show that dispatch records cannot serve as delay data while travel and handling measured from GPS are stable over time.

The closest work is the robust data-driven routing of Zhang, Zhang, Lim and Sim (*Operations Research*, 2021), which also reduces to deadlines brought forward by a buffer. Their bound holds over a Wasserstein ambiguity set inside a dedicated solver; ours is a finite-sample certificate that needs no change to the solver. Simheuristics rank solutions by simulated reliability within their own search; our certificate can be added on top of any of them. These differences are set out in Section 2.

I believe the paper fits the journal's interest in operations research for logistics and in reliability. The main claims were tested in three confirmatory studies whose hypotheses were fixed and publicly timestamped

before they were run, and all code, instances and records are openly available. The limitations are stated plainly. On the real routes, with the HGS solver, screening was more reliable than padding by 8.0 and 10.1 points under two of the three delay models, at the same number of vehicles; under the mildest model padding was already reliable and slightly ahead.

The manuscript is not under consideration elsewhere and has not been published. The public archive keeps the earlier drafts prepared from the same study, so that each can be rebuilt from the same records; none of them is under review or published. Generative AI (Claude and DeepSeek) was used for code, analysis, literature search and drafting, as disclosed in the manuscript; the author takes full responsibility for the content. There are no competing interests and no funding to declare.

I am an independent researcher without an institutional affiliation, so I use a personal e-mail address; my ORCID record (0009-0004-5079-7684) identifies me.

Suggested reviewers, none of whom has any connection with the author:

- Prof. Jan Fabian Ehmke, Department of Business Decisions and Analytics, University of Vienna (jan.ehmke@univie.ac.at), for service levels in routing with stochastic travel times;
- Dr. Ola Jabali, Operations Research Group, DEIB, Politecnico di Milano, for self-imposed time windows and routing under uncertainty;
- Prof. Angel A. Juan, Universitat Politècnica de València, for simheuristics.

Thank you for considering the manuscript.

Yours sincerely,

Suhan Ali
Independent researcher, Lahore, Pakistan
syedsuhanali9@gmail.com ORCID 0009-0004-5079-7684